VECTOR SPACES

Comprehensive Study Notes — CO2

RV University | First Year Computer Science

These notes comprehensively cover CO2: Vector Spaces, Subspaces, Span, Linear Independence, Basis, Dimension, Homogeneous Systems, and the four fundamental Matrix Spaces (Row, Column, Null, Left Null) along with Rank. Every topic has full axiomatic treatment, worked examples with proper matrix notation, comparison tables, and exam-ready practice questions with solutions.

Topics: Vector Space · Subspace · Span · Linear Independence/Dependence · Basis · Dimension · Homogeneous Eqns · Row/Column/Null/Left-Null Spaces · Rank

Module 1: Vector Spaces

1.1 Informal Idea

A **vector space** is a collection of objects (called *vectors*) where you can add any two and scale any one by a number, with the result always staying inside the collection. The formal definition pins down exactly what "behaves correctly" means via ten axioms.

1.2 Formal Definition

Definition: Vector Space

A **vector space** over a field F is a non-empty set V equipped with two binary operations — **vector addition** $(+)$ and **scalar multiplication** $(\cdot)$ — satisfying ten axioms for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $c, d \in F$.

The Ten Axioms

#	Name	Statement	Group
(a)	Closure under +	$\mathbf{u}+\mathbf{v} \in V$	Abelian Group w.r.t. +
(b)	Associativity of +	$(\mathbf{u}+\mathbf{v})+\mathbf{w} = \mathbf{u}+(\mathbf{v}+\mathbf{w})$	
(c)	Additive identity	$\exists\, \mathbf{0} \in V : \mathbf{u}+\mathbf{0} = \mathbf{u}$	
(d)	Additive inverse	$\forall\, \mathbf{u} \in V,\ \exists -\mathbf{u} \in V : \mathbf{u}+(-\mathbf{u}) = \mathbf{0}$	
(e)	Commutativity of +	$\mathbf{u}+\mathbf{v} = \mathbf{v}+\mathbf{u}$	
(1)	Closure under ·	$c\,\mathbf{u} \in V$	Scalar mult.
(2)	Distributive (scalar)	$(c+d)\mathbf{u} = c\mathbf{u}+d\mathbf{u}$	
(3)	Scalar identity	$1 \cdot \mathbf{u} = \mathbf{u}$	
(4)	Distributive (vector)	$c(\mathbf{u}+\mathbf{v}) = c\mathbf{u}+c\mathbf{v}$	
(5)	Associativity of ·	$(cd)\mathbf{u} = c(d\mathbf{u})$	

Abelian Group Structure

Axioms (a)–(e) say $(V,+)$ is an **Abelian (commutative) group**. Axioms (1)–(5) govern scalar multiplication. The additive identity $\mathbf{0}$ is the **zero vector**.

1.3 Consequences from the Axioms

Derived Properties

For any $\mathbf{u} \in V$, $c \in F$:

1. $0 \cdot \mathbf{u} = \mathbf{0}$ (scalar zero $\times$ any vector = zero vector)
2. $c \cdot \mathbf{0} = \mathbf{0}$ (any scalar $\times$ zero vector = zero vector)
3. $(-1)\mathbf{u} = -\mathbf{u}$ (additive inverse equals -1 times the vector)
4. $c\mathbf{u} = \mathbf{0} \Rightarrow c = 0$ or $\mathbf{u} = \mathbf{0}$
5. The zero vector $\mathbf{0}$ is unique; each $-\mathbf{u}$ is unique

Zero Vector vs Zero Scalar

$\mathbf{0}$ (bold) = zero *vector* $\in V$. 0 (plain) = zero *scalar* $\in F$. They live in completely different sets!

1.4 Standard Examples

Example 1: $\mathbb{R}^n$

All n-tuples with component-wise operations. For $\mathbb{R}^3$:

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{pmatrix}, \qquad c\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} ca_1 \\ ca_2 \\ ca_3 \end{pmatrix}$$

$\dim(\mathbb{R}^n) = n$.

Example 2: $M_{m\times n}$ — All $m \times n$ Matrices

Matrix addition and scalar multiplication satisfy all 10 axioms. For $M_{3\times 1}$:

$$a\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} + b\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} au_1 + bv_1 \\ au_2 + bv_2 \\ au_3 + bv_3 \end{pmatrix} \in M_{3\times 1} \quad \text{(closed)}$$

$\dim(M_{m\times n}) = mn$. Zero vector is the zero matrix.

Example 3: More Vector Spaces

Set V	Operations	Dim	Standard basis
$\mathbb{R}^n$	component-wise	n	$e_1, \ldots, e_n$
$M_{m\times n}$	matrix + and $c\cdot$	mn	E_{ij} matrices
$P_n(\mathbb{R})$ (poly. deg $\le n$)	poly. + and $c\cdot$	$n+1$	$1, x, x^2, \ldots, x^n$
$C[a,b]$ (cont. functions)	function + and $c\cdot$	∞	—
$\{\mathbf{0}\}$ (trivial)	trivial	0	$\emptyset$

Example 4: NOT Vector Spaces

- **Even integers**: $\frac{1}{2} \times 2 = 1 \notin$ even integers. Scalar mult. closure (1) fails.
- **Odd integers**: $1 + 1 = 2$ is not odd; closure (a) fails. Also $\mathbf{0} = 0$ is not odd; axiom (c) fails.
- **Polynomials with degree exactly n**: $x^2 + (-x^2) = 0$ has no degree, not in set. Axiom (c) fails.

Module 2: Subspaces

2.1 Definition

Definition: Subspace

A non-empty subset W of a vector space V is a **subspace** of V if W is itself a vector space under the *same* binary operations as V.

2.2 Subspace Test — Only 3 Checks

Subspace Test

$W \subseteq V$ is a subspace $\iff$ all three hold:

1. **Zero vector**: $\mathbf{0} \in W$
2. **Closed under addition**: $\mathbf{u}, \mathbf{v} \in W \Rightarrow \mathbf{u} + \mathbf{v} \in W$
3. **Closed under scalar mult.**: $\mathbf{u} \in W, c \in F \Rightarrow c\mathbf{u} \in W$

(Conditions 2 & 3 combine as: $a\mathbf{u} + b\mathbf{v} \in W$ for all $\mathbf{u}, \mathbf{v} \in W$, $a, b \in F$.)

Why only 3?

All other axioms are *inherited* from V automatically since $W \subseteq V$.

2.3 Worked Examples

Subspace: Lines Through Origin

$W = \{(x, y) : y = 2x\}$ in $\mathbb{R}^2$.
$\mathbf{0} = (0, 0)$: $0 = 2(0)$ ✓ $(x_1, 2x_1) + (x_2, 2x_2) = (x_1 + x_2, 2(x_1 + x_2))$ ✓ $c(x, 2x) = (cx, 2cx)$ ✓
$\therefore W$ is a subspace. But $y = 2x + 1$ is **not**: $\mathbf{0} \notin W$ since $0 \neq 2(0) + 1$.

Subspace: Matrices of a Specific Form

$$W = \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} : a, b \in \mathbb{R} \right\} \subseteq M_{2\times 2}$$

$$\mathbf{0} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \in W \ (a = b = 0) \checkmark$$

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix} + \begin{pmatrix} c & d \\ -d & c \end{pmatrix} = \begin{pmatrix} a+c & b+d \\ -(b+d) & a+c \end{pmatrix} \text{ (same form) } \checkmark$$

$$k \begin{pmatrix} a & b \\ -b & a \end{pmatrix} = \begin{pmatrix} ka & kb \\ -kb & ka \end{pmatrix} \checkmark \quad \therefore W \text{ is a subspace.}$$

Non-Subspace: Zero Missing

$W = \left\{ \begin{pmatrix} a & a+1 \\ 0 & b \end{pmatrix} \right\}$: Requires $a = 0$ and $a + 1 = 0$ simultaneously — impossible. $\therefore \mathbf{0} \notin W$, **not a subspace**.

Non-Subspace from Class Notes

Is $\left\{\begin{pmatrix} a & a+1 \\ 0 & b \end{pmatrix}\right\}$ a subspace of $M_{2\times 2}$? **No** — satisfies closure and associativity but the zero matrix does not exist in the set, so axiom (c) fails.

2.4 Trivial Subspaces

Every V has: $\{\mathbf{0}\}$ (smallest subspace) and V itself (largest subspace).

Module 3: Span and Spanning Sets

3.1 Linear Combination

Linear Combination

A **linear combination** of $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$ is: $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k$ where $c_1, \dots, c_k \in F$.

3.2 Span

Span

The **span** of $S = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is the set of *all* linear combinations:

$$\text{span}(S) = \{c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k : c_i \in F\}$$

S **spans** V if $\text{span}(S) = V$.

Span is Always a Subspace

$\text{span}(S)$ is always a subspace of V — the *smallest* subspace containing all of S.

3.3 Examples

Span Test in $\mathbb{R}^3$

Does $\mathbf{b} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ lie in the span of $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$?

$$\left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & 1 & 2 \end{array}\right) \xrightarrow{R_3 \to R_3 - R_1 - R_2} \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{array}\right)$$

Last row: $0 = 2$ — inconsistent. So $\mathbf{b} \notin \text{span}(\mathbf{v}_1, \mathbf{v}_2)$.

Matrix Span — $M_{2\times 3}$

Show $M_{2\times 3} = \text{span}(E_{11}, E_{12}, E_{13}, E_{21}, E_{22}, E_{23})$ where E_{ij} has 1 at position (i, j):

$$E_{12} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} = a_{11}E_{11} + a_{12}E_{12} + a_{13}E_{13} + a_{21}E_{21} + a_{22}E_{22} + a_{23}E_{23}$$

$\dim(M_{2\times 3}) = 6$.

Polynomial Span

$P_2(\mathbb{R})$ is spanned by $\{1, x, x^2\}$: any $p(x) = a_0 + a_1x + a_2x^2 = a_0 \cdot 1 + a_1 \cdot x + a_2 \cdot x^2$.

Module 4: Linear Independence and Dependence

4.1 Definitions

Linear Independence

$\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} \subset V$ is **linearly independent** if the only solution to $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$ is $c_1 = c_2 = \cdots = c_k = 0$.
No vector in the set can be expressed as a linear combination of the others.

Linear Dependence

Linearly dependent: scalars exist, *not all zero*, with $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$.
Equivalently: at least one vector is a linear combination of the others.

4.2 Test via Matrix Row Reduction

Form $A = [\mathbf{v}_1 \mid \cdots \mid \mathbf{v}_k]$, reduce to RREF. All pivot columns $\Rightarrow$ independent. Any free column $\Rightarrow$ dependent.

4.3 Worked Examples

Example 1 — Dependent

$\mathbf{u} = \begin{pmatrix}1\\0\\0\end{pmatrix}$, $\mathbf{v} = \begin{pmatrix}0\\1\\0\end{pmatrix}$, $\mathbf{w} = \begin{pmatrix}2\\3\\0\end{pmatrix}$: $\mathbf{w} = 2\mathbf{u} + 3\mathbf{v} \Rightarrow$ **linearly dependent**.

Example 2 — Independent (Standard Basis)

$$a\begin{pmatrix}1\\0\\0\end{pmatrix} + b\begin{pmatrix}0\\1\\0\end{pmatrix} + c\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}a\\b\\c\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix} \Rightarrow a = b = c = 0$$

Linearly independent.

Example 3 — Row Reduction Test

$\mathbf{v}_1 = \begin{pmatrix}-1\\1\\1\end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix}-1\\-1\\1\end{pmatrix}$, $\mathbf{v}_3 = \begin{pmatrix}1\\-1\\1\end{pmatrix}$:

$$\begin{pmatrix}-1 & -1 & 1\\ 1 & -1 & -1\\ 1 & 1 & 1\end{pmatrix} \xrightarrow[R_3+R_1]{R_2+R_1} \begin{pmatrix}-1 & -1 & 1\\ 0 & -2 & 0\\ 0 & 0 & 2\end{pmatrix}$$

Three pivots $\Rightarrow$ **linearly independent**.

Example 4 — From Class Notes

$$(a) = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}, \quad (b) = \left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}, \quad (c) = \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$$

- (a): two identical vectors $\Rightarrow$ **dependent**.
- (b): $\begin{pmatrix} -1 \\ 1 \end{pmatrix} = -1 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ (scalar multiple) $\Rightarrow$ **dependent**.
- (c): 3 vectors in $\mathbb{R}^2$ (dim$=2$), $k > \dim(V) \Rightarrow$ always **dependent**.

4.4 Key Theorems

Fundamental Theorems on Independence

1. Set containing $\mathbf{0} \Rightarrow$ always **dependent**.
2. $k > \dim(V) \Rightarrow$ always **dependent**.
3. $k < \dim(V) \Rightarrow$ **cannot span** V.
4. Exactly $n = \dim(V)$ vectors: independent $\Leftrightarrow$ spans $V \Leftrightarrow$ basis.
5. Two vectors are dependent $\Leftrightarrow$ one is a scalar multiple of the other.

Module 5: Basis and Dimension

5.1 Basis

Basis

$\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a **basis** for V if:

1. $\mathcal{B}$ **spans** V
2. $\mathcal{B}$ is **linearly independent**

Equivalently: every $\mathbf{v} \in V$ has a *unique* representation $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$.

Coordinates

The unique scalars c_i are the **coordinates** of $\mathbf{v}$ in basis $\mathcal{B}$. Different bases give different coordinates for the same vector.

5.2 Standard Bases

For $\mathbb{R}^3$:

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

For $M_{2\times 2}$ $(\dim = 4)$:

$$E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Space	Standard Basis	Dim
$\mathbb{R}^n$	$\{e_1, \dots, e_n\}$	n
$M_{m\times n}$	$\{E_{ij} : 1 \le i \le m, 1 \le j \le n\}$	mn
$P_n(\mathbb{R})$	$\{1, x, x^2, \dots, x^n\}$	$n+1$
$\{\mathbf{0}\}$	$\emptyset$	0

5.3 Dimension

Dimension

$\dim(V)$ = number of vectors in any basis for V. All bases of V have the same number of elements.

Dimension Theorems

Let $\dim(V) = n$ and $S \subset V$ have $|S|$ vectors.

1. $|S| > n \Rightarrow S$ is dependent.
2. $|S| < n \Rightarrow S$ cannot span V.
3. $|S| = n$: independent $\Leftrightarrow$ spans V $\Leftrightarrow$ basis.

5.4 Worked Examples

Is S a Basis of $M_{2\times2}$?

$\dim(M_{2\times2}) = 4$. $S = \left\{ \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right\}$ has $|S| = 3 < 4$.
A set of 3 vectors cannot span a 4-dimensional space. $\therefore$ S is **not a basis** of $M_{2\times2}$.

Multiple Bases for the Same Space (from class)

For $W = \left\{ \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} : a, b \in \mathbb{R} \right\}$, all valid bases:

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}, \quad \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} \right\}, \quad \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} \right\}$$

Each has 2 vectors $\Rightarrow \dim(W) = 2$. The first is the special orthonormal basis.

Module 6: Homogeneous Systems

Homogeneous System

$A\mathbf{x} = \mathbf{0}$ (zero right-hand side) is **homogeneous**.

Key Properties

1. Always **consistent**: $\mathbf{x} = \mathbf{0}$ is always the trivial solution.
2. Non-trivial solution $\Leftrightarrow$ $\text{rank}(A) < n$ $\Leftrightarrow$ at least one free variable.
3. #rows < #columns: always non-trivial solutions.
4. Solution set is a **subspace** of $\mathbb{R}^n$ (the null space).

6.1 General Solution via Null Space

If $\mathbf{u}$ is the general solution to $A\mathbf{x} = \mathbf{0}$ and $\mathbf{v}$ is any particular solution to $A\mathbf{x} = \mathbf{b}$:

$$\mathbf{x}_{\text{general}} = \mathbf{v}_{\text{particular}} + \mathbf{u}_{\text{homogeneous}}$$

Proof: $A(\mathbf{v} + \mathbf{u}) = A\mathbf{v} + A\mathbf{u} = \mathbf{b} + \mathbf{0} = \mathbf{b}$ ✓

Worked Example

Solve: $2x_1 + 2x_2 + x_3 = 0,\ \ 2x_1 - 2x_2 - x_3 = 1$

$$\left(\begin{array}{ccc|c} 2 & 2 & 1 & 0 \\ 2 & -2 & -1 & 1 \end{array}\right) \xrightarrow{R_2 - R_1} \left(\begin{array}{ccc|c} 2 & 2 & 1 & 0 \\ 0 & -4 & -2 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{1}{4} \\ 0 & 1 & \frac{1}{2} & -\frac{1}{4} \end{array}\right)$$

x_3 free. Null(A): $u_1 = 0$, $u_2 = -\frac{1}{2}u_3$:

$$\text{Null}(A) = \text{span}\left\{\begin{pmatrix} 0 \\ -\frac{1}{2} \\ 1 \end{pmatrix}\right\}, \quad \mathbf{v}_{\text{part}} = \begin{pmatrix} \frac{1}{4} \\ -\frac{1}{4} \\ 0 \end{pmatrix} \quad \Rightarrow \quad \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \alpha \begin{pmatrix} 0 \\ -\frac{1}{2} \\ 1 \end{pmatrix} + \begin{pmatrix} \frac{1}{4} \\ -\frac{1}{4} \\ 0 \end{pmatrix}, \ \alpha \in \mathbb{R}$$

Module 7: The Four Fundamental Matrix Spaces

7.1 Overview

For any $m \times n$ matrix A with rank r, there are four fundamental subspaces:

Space	Symbol	Definition	Lives in	Dim
Row space	$\text{Row}(A)$	Span of rows of A	$\mathbb{R}^n$	r
Column space	$\text{Col}(A)$	Span of columns of A	$\mathbb{R}^m$	r
Null space	$\text{Null}(A)$	Solutions of $A\mathbf{x} = \mathbf{0}$	$\mathbb{R}^n$	$n - r$
Left null space	$\text{Null}(A^T)$	Solutions of $A^T\mathbf{y} = \mathbf{0}$	$\mathbb{R}^m$	$m - r$

Rank-Nullity + Orthogonality

$$\begin{aligned}
\dim(\text{Col}(A)) = \dim(\text{Row}(A)) &= r = \text{rank}(A) \\
\dim(\text{Null}(A)) = n - r, &\qquad \dim(\text{Null}(A^T)) = m - r \\
\text{Row}(A) \perp \text{Null}(A) &\quad \text{(orthogonal complements in } \mathbb{R}^n) \\
\text{Col}(A) \perp \text{Null}(A^T) &\quad \text{(orthogonal complements in } \mathbb{R}^m)
\end{aligned}$$

7.2 Column Space — Col(A)

Column Space

$\text{Col}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\}$ = span of columns of A, a subspace of $\mathbb{R}^m$.
It is the set of all $\mathbf{b}$ for which $A\mathbf{x} = \mathbf{b}$ is *consistent*.

Algorithm: Row-reduce A to RREF → identify pivot column positions → take those columns from **original** A.

Column Space — From Class Notes

$$A = \begin{pmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{pmatrix}$$

$$\text{rref}(A) = \begin{pmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Pivot cols: 1 and 3 ($C_2 \to 2 \times C_1$, dependent). Basis from original A:

$$\text{Col}(A) = \text{span}\left\{ \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ 5 \end{pmatrix} \right\}, \quad \dim(\text{Col}(A)) = 2$$

Common Mistake

Use the **original** columns of A, NOT columns of RREF, for the $\text{Col}(A)$ basis. Row operations change the column space! Only the pivot *positions* come from RREF.

7.3 Null Space — $\mathrm{Null}(A)$

Null Space / Kernel

$\mathrm{Null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$. Always a subspace of $\mathbb{R}^n$. $\dim(\mathrm{Null}(A)) = \#\text{free variables} = n - \mathrm{rank}(A)$.

Algorithm: RREF of A → express basic vars in terms of free vars → write $\mathbf{x}$ as linear combination (one basis vector per free variable).

Null Space — 5-variable Example

$A = \begin{pmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{pmatrix}$, $\mathrm{rref}(A) = \begin{pmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$
Pivots: cols 1,3 ⇒ x_1, x_3 basic; x_2, x_4, x_5 free. $x_1 = 2x_2 + x_4 - 3x_5$, $x_3 = -2x_4 + 2x_5$:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = x_2 \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{pmatrix} + x_5 \begin{pmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{pmatrix}$$

$$\mathrm{Null}(A) = \mathrm{span}\left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{pmatrix} \right\}, \quad \dim = 3$$

Check: $\dim \mathrm{Col} + \dim \mathrm{Null} = 2 + 3 = 5 = n$ ✓

Null Space — 3×4 (from class)

$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$, $\mathrm{rref}(A) = \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix}$
x_4 free; $x_1 = -2x_4$, $x_2 = x_4$, $x_3 = x_4$:

$$\mathrm{Null}(A) = \mathrm{span}\left\{ \begin{pmatrix} -2 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

7.4 Row Space — $\mathrm{Row}(A)$

Row Space

$\mathrm{Row}(A) = \mathrm{Col}(A^T)$ = span of row vectors of A. Subspace of $\mathbb{R}^n$.

Algorithm: Row-reduce A to REF/RREF. The **non-zero rows** of RREF form a basis for $\mathrm{Row}(A)$.

Row vs Col — Key Difference

$\mathrm{Row}(A)$ basis: rows of **RREF of** A. $\mathrm{Col}(A)$ basis: pivot columns of **original** A.
Row operations preserve row space. They do NOT preserve column space.

Row Space — Large Matrix

$$A = \begin{pmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{pmatrix}$$

Take A^T and find $\text{Col}(A^T)$. After row-reducing A^T, pivot positions give us rows 1,2,4 of original A as the basis:

$$\text{Row}(A) = \text{span}\left\{ \begin{pmatrix} 1 \\ -2 \\ 0 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ -5 \\ -3 \\ -2 \\ 6 \end{pmatrix}, \begin{pmatrix} 2 \\ 6 \\ 18 \\ 8 \\ 6 \end{pmatrix} \right\}, \quad \dim = 3$$

7.5 Left Null Space — Null(A^T)

Left Null Space

$\text{Null}(A^T) = \{\mathbf{y} \in \mathbb{R}^m : A^T\mathbf{y} = \mathbf{0}\}$. Equivalently $\mathbf{y}^T A = \mathbf{0}^T$ ("left"). Subspace of $\mathbb{R}^m$. $\dim = m - r$.

Algorithm: RREF of A^T, solve $A^T\mathbf{y} = \mathbf{0}$.

7.6 Grand Worked Example — All Four Spaces

Complete Example: 4×5 Matrix (from class notes)

$$A = \begin{pmatrix} 2 & 3 & -1 & 1 & 2 \\ -1 & -1 & 0 & -1 & 1 \\ 1 & 2 & -1 & 1 & 1 \\ 1 & -2 & 3 & 1 & -3 \end{pmatrix}, \quad m = 4, n = 5$$

$$\text{rref}(A) = \begin{pmatrix} 1 & 0 & 1 & 0 & -1 \\ 0 & 1 & -1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Pivot cols: 1,2,4. Rank $r = 3$.
Col(A) — cols 1,2,4 of original A:

$$\text{Col}(A) = \text{span}\left\{ \begin{pmatrix} 2 \\ -1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ -1 \\ 2 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \\ 1 \end{pmatrix} \right\}, \quad \dim = 3$$

Row(A) — non-zero rows of RREF:

$$\text{Row}(A) = \text{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ -2 \end{pmatrix} \right\}, \quad \dim = 3$$

Null(A) — free: x_3, x_5; $x_1 = -x_3 + x_5$, $x_2 = x_3 - 2x_5$, $x_4 = 2x_5$:

$$\text{Null}(A) = \text{span}\left\{\begin{pmatrix}-1\\1\\1\\0\\0\end{pmatrix}, \begin{pmatrix}1\\-2\\0\\2\\1\end{pmatrix}\right\}, \quad \dim = 5 - 3 = 2$$

Left Null(A) — solve $A^T\mathbf{y} = \mathbf{0}$ (one free variable):

$$\text{Null}(A^T) = \text{span}\left\{\begin{pmatrix}-2\\2\\5\\1\end{pmatrix}\right\}, \quad \dim = 4 - 3 = 1$$

Checks: $3+2 = 5 = n$ ✓ $3+1 = 4 = m$ ✓ Row ⊥ Null: $(1,0,1,0,-1)\cdot(-1,1,1,0,0)^T = -1+0+1+0+0 = 0$ ✓

Module 8: Rank

Rank

$\text{rank}(A) = \dim(\text{Col}(A)) = \dim(\text{Row}(A)) = \#$ pivot columns in RREF.

Rank-Nullity Theorem

For $m \times n$ matrix A with rank r:

$$r + (n - r) = n \quad \text{and} \quad r + (m - r) = m$$

rank + nullity = #columns. #lin. indep. rows = #lin. indep. columns = rank.

Space	How to find basis	Ambient	Dim	Orthog. partner
$\text{Col}(A)$	Pivot cols of RREF → orig. A	$\mathbb{R}^m$	r	$\text{Null}(A^T)$
$\text{Row}(A)$	Non-zero rows of RREF	$\mathbb{R}^n$	r	$\text{Null}(A)$
$\text{Null}(A)$	Solve $A\mathbf{x} = \mathbf{0}$; 1 vec per free var	$\mathbb{R}^n$	$n - r$	$\text{Row}(A)$
$\text{Null}(A^T)$	Solve $A^T\mathbf{y} = \mathbf{0}$	$\mathbb{R}^m$	$m - r$	$\text{Col}(A)$

Module 9: Exam Preparation

9.1 Top 20 Most Testable Facts

1. Vector space: **10 axioms** (5 for abelian group, 5 for scalar mult).
2. The zero vector $\mathbf{0}$ is **unique** in any vector space.
3. Subspace: only **3 checks** needed ($\mathbf{0} \in W$, closed under $+$, closed under $\cdot$).
4. span(S) is always a **subspace** — the smallest containing S.

5. Independent: $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$ has only trivial solution.
6. Set containing $\mathbf{0}$ $\Rightarrow$ always **dependent**.
7. $k > \dim(V)$ $\Rightarrow$ always **dependent**.
8. $k < \dim(V)$ $\Rightarrow$ **cannot span** V.
9. Basis = spans + independent; every vector has a unique expansion.
10. All bases of V have the **same number** of vectors.
11. $\dim(M_{m\times n}) = mn$; $\dim(P_n) = n+1$; $\dim(\mathbb{R}^n) = n$.
12. Homogeneous $A\mathbf{x} = \mathbf{0}$: **always consistent**.
13. Non-trivial solution $\Leftrightarrow$ $\text{rank}(A) < n$.
14. General soln of $A\mathbf{x} = \mathbf{b}$ = particular + null space solution.
15. $\text{Col}(A)$ basis: use **original** A columns at pivot positions.
16. $\text{Row}(A)$ basis: **non-zero rows** of RREF.
17. $\text{Null}(A)$ basis: one vector per **free variable**.
18. $\text{Null}(A^T)$: solve $A^T\mathbf{y} = \mathbf{0}$.
19. Rank-Nullity: $r + (n-r) = n$ and $r + (m-r) = m$.
20. $\text{Row}(A) \perp \text{Null}(A)$; $\text{Col}(A) \perp \text{Null}(A^T)$.

9.2 Common Mistakes

Mistakes to Avoid

- Using RREF columns (not original A) for $\text{Col}(A)$ basis.
- Forgetting to check $\mathbf{0} \in W$ when testing subspaces.
- Thinking a basis is unique — infinitely many exist.
- Claiming k vectors span $\mathbb{R}^n$ when $k < n$.
- Confusing $\text{Null}(A) \subset \mathbb{R}^n$ with $\text{Null}(A^T) \subset \mathbb{R}^m$.
- Not doing the Rank-Nullity sanity check after finding spaces.
- Thinking row operations preserve the column space (they do NOT).

9.3 Practice MCQs

Q1. Which is always a subspace of $\mathbb{R}^3$?
(A) $x+y+z=1$ **(B)** $x+y+z=0$ (C) all entries > 0 (D) all unit vectors

Q2. $\dim(V) = 5$, $|S| = 7$, $S \subset V$. Then S is:
(A) linearly dependent (B) a basis (C) independent (D) spanning

Q3. For $A_{3\times 7}$ with rank 3, $\dim(\text{Null}(A)) =$
(A) 3 (B) 7 **(C) 4** (D) 0

Q4. $\text{Null}(A^T)$ lives in:
(A) $\mathbb{R}^m$ (B) $\mathbb{R}^n$ (C) $\mathbb{R}^r$ (D) $\mathbb{R}^{mn}$

Q5. $A\mathbf{x} = \mathbf{0}$ has non-trivial solution iff:
(A) A invertible **(B) rank**$(A) < n$ (C) $m > n$ (D) $A = 0$

Q6. Which pair are orthogonal complements?
(A) $\text{Col}(A)$ and $\text{Row}(A)$ **(B) Null**(A) **and Row**(A) (C) $\text{Col}(A)$ and $\text{Null}(A)$

Q7. $\dim(M_{2\times 3}) =$
(A) 2 (B) 3 **(C) 6** (D) 5

Q8. A basis of $P_3(\mathbb{R})$ has how many elements?
(A) 3 **(B) 4** (C) 5 (D) infinite

9.4 Practice Descriptive Questions

1. Verify all 10 vector space axioms for $\mathbb{R}^3$.

2. Is $W = \{(x, y, z) : x + y + z = 0\}$ a subspace of $\mathbb{R}^3$? Find basis and dimension.

3. Are $\left\{ \begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}2\\1\\0\end{pmatrix}, \begin{pmatrix}0\\3\\6\end{pmatrix} \right\}$ linearly independent?

4. Find all four fundamental spaces for $A = \begin{pmatrix} 2 & 3 & -1 & 1 & 2 \\ -1 & -1 & 0 & -1 & 1 \\ 1 & 2 & -1 & 1 & 1 \\ 1 & -2 & 3 & 1 & -3 \end{pmatrix}$.

5. Verify Rank-Nullity for the matrix in Q4.

6. Solve and express as particular + null-space: $A = \begin{pmatrix} 2 & 2 & 1 \\ 2 & -2 & -1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix}0\\1\end{pmatrix}$.

7. Prove $\text{Row}(A) \perp \text{Null}(A)$.

8. Show $M_{2\times3} = \text{span}(E_{11}, E_{12}, E_{13}, E_{21}, E_{22}, E_{23})$.

Cheat Sheet — Quick Reference

Vector Space — 10 Axioms

Abelian group w.r.t. +:

(a) Closed: $\mathbf{u}+\mathbf{v} \in V$
(b) Assoc: $(\mathbf{u}+\mathbf{v})+\mathbf{w} = \mathbf{u}+(\mathbf{v}+\mathbf{w})$
(c) Identity: $\exists \mathbf{0}$, $\mathbf{u}+\mathbf{0}=\mathbf{u}$
(d) Inverse: $\exists -\mathbf{u}$, $\mathbf{u}+(-\mathbf{u})=\mathbf{0}$
(e) Comm: $\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$

Scalar multiplication:

(1) $c\mathbf{u} \in V$
(2) $(c+d)\mathbf{u} = c\mathbf{u}+d\mathbf{u}$
(3) $1\mathbf{u}=\mathbf{u}$
(4) $c(\mathbf{u}+\mathbf{v}) = c\mathbf{u}+c\mathbf{v}$
(5) $(cd)\mathbf{u} = c(d\mathbf{u})$

Subspace Test (3 Conditions)

$\mathbf{0} \in W$ $\quad \mathbf{u},\mathbf{v} \in W \Rightarrow \mathbf{u}+\mathbf{v} \in W$ $\quad \mathbf{u} \in W \Rightarrow c\mathbf{u} \in W$

Linear Independence Test

$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$ has only $c_i = 0$
$\Leftrightarrow$ all columns of $[\mathbf{v}_1|\cdots|\mathbf{v}_k]$ are pivots.

Basis = Independent + Spanning

Every $\mathbf{v} \in V$ has a *unique* expansion in $\mathcal{B}$.

Standard Dimensions

$\dim(\mathbb{R}^n) = n$, $\quad \dim(M_{mn}) = mn$, $\quad \dim(P_n) = n+1$

Key Mnemonics

Col basis = **col**umns of **orig**inal A.
Row basis = **rows** of RREF.
Null: 1 basis vector per **free** variable.
$r + \text{nullity} = n$ — always sanity-check!

Four Spaces ($m \times n$, rank r)

- $\text{Col}(A) \subset \mathbb{R}^m$; $\dim = r$; orig. pivot cols
- $\text{Row}(A) \subset \mathbb{R}^n$; $\dim = r$; non-zero rows of RREF
- $\text{Null}(A) \subset \mathbb{R}^n$; $\dim = n-r$; solve $A\mathbf{x}=\mathbf{0}$
- $\text{Null}(A^T) \subset \mathbb{R}^m$; $\dim = m-r$; solve $A^T\mathbf{y}=\mathbf{0}$

Rank-Nullity

$r + (n-r) = n$ (cols) $\quad r + (m-r) = m$ (rows)

Orthogonality

$\text{Row}(A) \perp \text{Null}(A)$ in $\mathbb{R}^n$
$\text{Col}(A) \perp \text{Null}(A^T)$ in $\mathbb{R}^m$

General Solution to $A\mathbf{x} = \mathbf{b}$

$\mathbf{x} = \mathbf{v}_{\text{part}} + \mathbf{u}_{\text{hom}}$
$A\mathbf{v}_{\text{part}} = \mathbf{b}$, $\quad A\mathbf{u}_{\text{hom}} = \mathbf{0}$

Big Picture Diagram

Space	Lives in	Dim
$\text{Col}(A^T) = \text{Row}(A)$	$\mathbb{R}^n$	r
$\text{Col}(A)$	$\mathbb{R}^m$	r
$\text{Null}(A) : A\mathbf{x}=\mathbf{0}$	$\mathbb{R}^n$	$n-r$
$\text{Null}(A^T) : A^T\mathbf{y}=\mathbf{0}$	$\mathbb{R}^m$	$m-r$

References

Lay et al., *Linear Algebra*, 5th ed., Pearson 2016.
Anton & Rorres, *Elem. Linear Algebra*, 11th ed.
CO2 Class Notes, RV University.